



SVKM'S NMIMS
MUKESH PATEL SCHOOL OF TECHNOLOGY MANAGEMENT & ENGINEERING/
SCHOOL OF TECHNOLOGY MANAGEMENT

Academic Year: 2023-2024

Program/s: B. Tech

Year: II Semester: IV

Stream/s : CSE (Cybersecurity)

Subject: Introduction to Cryptography

Time: 3 hrs (10:00 am. To 01:00 pm.)

Date: 27 June 2024

No. of Pages: 03

Marks: 100

Re-Examination (2021-22, 2022-23, 2023-24)

Instructions: Candidates should read carefully the instructions printed on the question paper and on the cover of the Answer Book, which is provided for their use.

- 1) Question No. 1 is compulsory.
- 2) Out of remaining questions, attempt any 4 questions.
- 3) **In all 5 questions to be attempted.**
- 4) All questions carry equal marks.
- 5) **Answer to each new question to be started on a fresh page.**
- 6) **Figures in brackets on the right hand side indicate full marks.**
- 7) **Assume Suitable data if necessary.**

Q 1		Solve:	
CO - 1; BL- 3	a.	The following text has been obtained by first encoding the plaintext by using Vigenère Cipher with the key "RIVER" and then encoding the result again by Permutation Cipher with the key. $\pi = \begin{matrix} 1 & 2 & 3 & 4 & 5 \\ 5 & 4 & 2 & 3 & 1 \end{matrix}$ Decrypt the text: YXTJTVVBJJ	[05]
CO - 1; BL- 3	b.	Decrypt the following ciphertext obtained from the Autokey Cipher by using the key $k = 12$ OJHRD	[05]
CO - 1; BL- 3	c.	$\begin{pmatrix} 7 & 2 \\ 3 & 5 \end{pmatrix}$ Find the inverse of the following matrix in Z_{26}:	[05]
CO - 1; BL- 3	d.	An operator of a Vigenère Cipher encryption machine has nothing to do and encrypts the plaintext zzz . . . (30 times), that is, the plaintext consists of the same letter x thirty times, with the key "bored". The operator obtained six blocks of same ciphertext. Find the ciphertext block	[05]

Q 2 CO - 2; BL- 3	a.	Suppose that Bob has decided to set up a RSA system with $n = 55 = p * q$, where p and q are prime numbers. If $p < q$, then find p and q and compute $\phi(55)$.	[10]																								
Q 2 CO - 2; BL- 3	b.	If $(n, b) = (55, 3)$ is a public key for the RSA system in the previous problem <i>Q.2a</i> , then its corresponding private key is of the form (p, q, a) . Find the value of a .	[10]																								
Q 3 CO - 3; BL- 3	a.	Alice and Bob agree to use the prime $p = 19$ and the base $g = 3$ for a Diffie-Hellman key exchange. Alice sends the value $A = 7$. Bob selects his secret exponent as $b = 8$. Find the value of B that Bob sends to Alice.	[10]																								
Q 3 CO - 3; BL- 3	b.	Find the secret shared key computed by Bob and Alice in the previous problem <i>Q.3a</i> . Suppose you guessed correctly that Alice's secret exponent is slightly lower than Bob's exponent, find Alice's secret exponent a in the previous problem <i>Q.3a</i> .	[10]																								
Q 4 CO - 2; BL- 3	a.	In order to use El Gamal cryptosystem, Alice selects the prime number $p = 29$ and $g = 2$. She selects her private key $a = 10$ and computes A . Find the value of A . <i>Remember that your answer should be an integer between 0 and 29.</i>	[10]																								
Q 4 CO - 2; BL- 3	b.	Upon receiving A from Alice as in the previous problem <i>Q.4a</i> , Bob selects his secret random element $k = 8$ in order to send the message $m = 6$. He computes the ciphertext (y_1, y_2) . Compute the value of y_1 and y_2 .	[10]																								
Q 5 CO - 1; BL- 3	a.	Square-and-Multiply Algorithm (x, b, n): - Step 1: $z \leftarrow 1$ - Step 2: For $i = \ell - 1$ down to 0, do - Step 2.1: $z \leftarrow z^2 \pmod{n}$ - Step 2.2: If $b_i = 1$, then $z \leftarrow z \times x \pmod{n}$ - Step 3: Return z Compute $10^9 \pmod{17}$ by filling out the following table, where b_i's are binary bits of 9. Note: No marks will be awarded if the table is left blank and no marks will be awarded for directly computing $10^9 \pmod{17}$ without using square-and-multiply algorithm. <table border="1" style="margin-left: auto; margin-right: auto;"> <tr> <th>i</th><th>b_i</th><th></th><th>z</th></tr> <tr> <td></td><td></td><td></td><td>1</td></tr> <tr> <td>3</td><td></td><td></td><td></td></tr> <tr> <td>2</td><td></td><td></td><td></td></tr> <tr> <td>1</td><td></td><td></td><td></td></tr> <tr> <td>0</td><td></td><td></td><td></td></tr> </table>	i	b_i		z				1	3				2				1				0				[10]
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Q 5 CO - 1; BL- 3	b.	Suppose that the Affine Cipher with the encryption function $e(x) = 3x + 10 \pmod{26}$ is used to encrypt a plaintext. Find the decryption function $d(y) = cy + d \pmod{26}$. If the Affine Cipher yields the following ciphertext, find the corresponding plaintext : NAXT	[10]																								

Q 6 CO - 2; BL- 3	a.	Consider the finite field $F = \mathbb{Z}_2[x] / (x^4 + x + 1)$ for problems Q.6 and Q.7. Compute $(x^2 + 1)(x^3 + x^2 + x)$ in the field F and represent it in binary bits	[05]
Q 6 CO - 1; BL- 3	b.	Consider the Substitute nibble $\pi_S: \{0, 1\}^4 \rightarrow \{0, 1\}^4$ that is part of the components of the Simplified AES, which is given as follows: - The permutation π_S incorporates operations in the finite field F. - Identify the binary string $a_3a_2a_1a_0$ of length four with the field element $y = a_3x^3 + a_2x^2 + a_1x + a_0$ and let $y^{-1} = b_3x^3 + b_2x^2 + b_1x + b_0$ be the inverse of y in F (with $0^{-1} = 0$). - Then, compute $\begin{pmatrix} c_3 \\ c_2 \\ c_1 \\ c_0 \end{pmatrix} = \begin{pmatrix} 1 & 0 & 1 & 1 \\ 1 & 1 & 0 & 1 \\ 1 & 1 & 1 & 0 \\ 0 & 1 & 1 & 1 \end{pmatrix} \begin{pmatrix} b_3 \\ b_2 \\ b_1 \\ b_0 \end{pmatrix} \oplus \begin{pmatrix} 1 \\ 0 \\ 0 \\ 1 \end{pmatrix}$ - Define $\pi_S(a_3a_2a_1a_0) = c_3c_2c_1c_0$ Using the above description and the fact that the inverse of E is 3, compute $\pi_S(E)$ in hexadecimal form.	[15]
Q 7 CO - 1; BL- 3	a.	Consider the Mix Column that is part of the components of the Simplified AES, which is given as the following matrix multiplication in the field $F = \mathbb{Z}_2[x] / (x^4 + x + 1)$: $\begin{bmatrix} 1 & 4 \\ 4 & 1 \end{bmatrix} \begin{bmatrix} c_{0,0} & c_{0,1} \\ c_{1,0} & c_{1,1} \end{bmatrix} = \begin{bmatrix} d_{0,0} & d_{0,1} \\ d_{1,0} & d_{1,1} \end{bmatrix}$ If $c_{0,0} = 4$, $c_{0,1} = D$, $c_{1,0} = A$, and $c_{1,1} = 9$ in hexadecimal notation, find the resulting matrix after the Mix Column. Write $d_{1,0}$ and $d_{0,1}$ of your matrix in hexadecimal notation	[10]
Q 7 CO - 1; BL- 3	b.	Recall that a key k is called an involutory key if the encryption function e_k is the same as the decryption function d_k . Find the number of involutory keys in the permutation cipher with $m = 3$.	[10]

Alphabet Codings

A	B	C	D	E	F	G	H	I	J	K	L	M
0	1	2	3	4	5	6	7	8	9	10	11	12

N	O	P	Q	R	S	T	U	V	W	X	Y	Z
13	14	15	16	17	18	19	20	21	22	23	24	25

Inverses in $\mathbb{Z}_{26}$

a	1	3	5	7	11	17	25
a^{-1}	1	9	21	15	19	23	25